

Phases of the Moon

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Over about a month the Moon seems to change shape: **new**, waxing crescent, first quarter, waxing gibbous, **full**, waning gibbous, last quarter, waning crescent. **Waxing** means the light is growing; **waning** means the light is shrinking. The Moon does not really change shape—we see different amounts of its lit half as it orbits Earth.

Round 1 — Retrieve It

1. About how long does it take the Moon to go through all its phases?

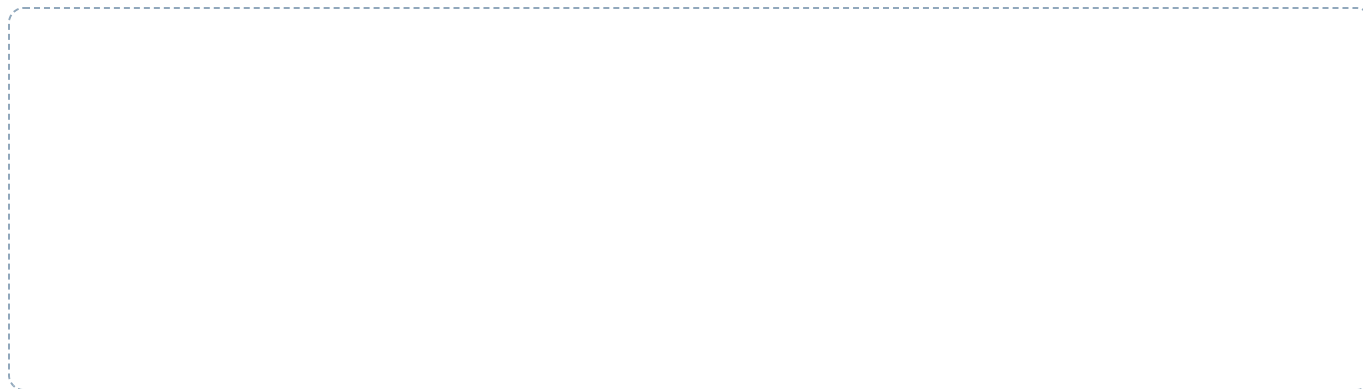
2. What do we call the phase when we cannot see the lit side of the Moon at all?

3. What does it mean when the Moon is waxing?

4. Does the Moon really change its shape? Explain in one sentence.

Round 2 — Sketch It

Draw four Moon phases in order: new, first quarter, full, and last quarter. Label each one and shade the dark part.



Phases of the Moon — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Idea

1. Tonight you see a thin sliver of Moon that is getting bigger each night. Is it waxing or waning, and how do you know?

2. You see a full Moon tonight. What phase will come a few nights later as the light shrinks?

3. Put these phases in order starting from new Moon: full, first quarter, new. Then tell what comes next.

Round 4 — Reason from Evidence

A friend says, "The Moon shrinks and then grows back each month." Using what you know, explain why the Moon looks like it changes shape even though it does not.

Phases of the Moon — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain the difference between waxing and waning. Give one phase that is waxing and one that is waning.

C**Connect**

Connect the Moon's orbit to what we see. Why do we see different amounts of the lit half as the Moon orbits Earth?

E**Extend**

Plan a "Moon journal." What would you draw and write each night for two weeks to show the phases changing?

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Phases of the Moon

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. About one month (about 29 to 30 days).
2. New Moon.
3. The lit part we can see is growing (getting larger each night).
4. No. It looks different because we see changing amounts of its lit half as it orbits Earth.

Sketch It — look for

Four phases in order and labeled: new (dark), first quarter (half lit), full (fully lit), last quarter (other half lit), with the dark part shaded.

Apply It

1. Waxing; the lit part is getting bigger each night, which means the light is growing.
2. Waning gibbous (the light shrinks after full Moon).
3. New, first quarter, full; next comes the waning gibbous (light begins to shrink).

Reason from Evidence — look for

The Moon does not change shape. As it orbits Earth, the Sun lights up half of it, and we see different amounts of that lit half from Earth, so it seems to grow and shrink.

ACE — Extend look-for

A plan to observe and draw the Moon at the same time each night, noting the date and whether it is waxing or waning, showing the shape changing over two weeks.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.